

**Indo-Islamic Painting in the Unpublished Miniatures in the Manuscript
"AJA'IB AL-SANA'I" Preserved in the British Library (Or.13718), Malwa
Region in the Khalji Dynasty Era**

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An Artistic, Archaeological, and Comparative Study

التصوير الهندي الإسلامي في مجموعة لوحات غير منشورة من مخطوط «عجائب الصنائع» المحفوظ في
المكتبة البريطانية بلندن (Or.13718)، إقليم مالوه عصر الأسرة الخلجية، دراسة أثرية فنية مقارنة

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ABSTRACT

This paper investigates and publishes a collection of miniatures in the manuscript "*Aja'ib al-Sana'i*" for the first time. It represents the Persian copy of the manuscript "*al-hiyal al-handasiyah*" by al-Jazari, which was copied and decorated in **Mandu**, the capital of the **Malwa** in India during the rule of **Nasser Shaha Khalji** (906-916AH/1500-1510AD). The translated copy of Al-Jazari's manuscript of "*al-hiyal al-handasiyah*" from Arabic into Persian, with its decoration and presentation named "*Aja'ib Al-Sana'i*" is one of the manuscripts with new topics of the Islamic painting in India during the era of the sultans. It is distinctive, especially in light of the dominance of Iranian artistic influences, on the one hand and the prevalence of artistic heritage among artists, on the other. The manuscript's illustrations provide new archaeological evidence of scientific and artistic connections between India and the Islamic states of Egypt, the Levant, and the Arabian Peninsula. Concerning the artistic style, which was prevalent in Mandu during the Khalji era, it has been studied in terms of the drawings of people, the clarity of social classes, the choice of animals, mythical creatures, and birds that are the artist's favorites, in addition to the colors, which were largely limited, and the extent to which the Indian artistic heritage was evident. The research illustrates the transfer of geometric illustrations from the original Arabic version of the manuscript and discusses the core differences between the Indian, Persian, and Arabic versions in terms of the design of the mechanical devices and the method of drawing live models, whether human or animal, inside the devices and the extent of the artist's success. In addition, just as the name changed, there are many core differences between the Indian version under study and the Persian and Arabic versions, especially the lack of commitment to realism that characterized al-Jazari's illustrated copies and the different approach in the geometric illustrations of the famous device from the manuscript of al-hiyal, which is the elephant clock. The descriptive, analytical study refutes the number of participants implementing the illustrations and provides evidence through the implemented artistic method.

KEYWORDS: Indo- Islamic Painting, Manuscripts, Persian, Watches, Wonders.

ملخص

يتناول هذا البحث وينشر لأول مرة مجموعة من لوحات مخطوط "عجائب الصنائع"، وهو ترجمة فارسية لمخطوطة "الحيل الهندسية" التي ألفها "ابن الرذاذ الجزري"، نُسخَت المخطوطة وزُوِّقَت باللوحات في مدينة ماندو، عاصمة إقليم مالوا في الهند، في عهد السلطان "ناصر شاه خلجي" (906-916هـ/1500-1510م). وتعد هذه المخطوطة المزوقة "عجائب الصنائع" من المخطوطات التي طرحت موضوعات جديدة في التصوير الإسلامي في الهند في عهد السلاطين؛

لا سيما في ظل هيمنة التأثيرات الفنية الإيرانية من جهة، وانتشار الموروث الفني بين الفنانين من جهة أخرى. تُقدم رسوم المخطوطة شواهد أثرية جديدة على الروابط العلمية والفنية بين الهند والدول الإسلامية في مصر والشام وشبه الجزيرة العربية. أما فيما يتعلق بالأسلوب الفني السائد في ماندو خلال العصر الخلجي، فقد تمت دراسته من حيث رسومات الأشخاص، ووضوح الطبقات الاجتماعية، واختيار الحيوانات والكائنات الخرافية والطيور التي كانت مفضلة لدى الفنان، بالإضافة إلى الألوان التي كانت محدودة إلى حد كبير، ومدى وضوح الموروث الفني الهندي. يوضح البحث نقل الرسوم الهندسية من النسخة العربية الأصلية للمخطوطة، ويناقش الاختلافات الجوهرية بين النسخ الهندية والفارسية والعربية من حيث تصميم الأجهزة الميكانيكية وطريقة رسم النماذج الحية، بشرية كانت أم حيوانية، داخل الأجهزة ومدى نجاح الفنان. إضافةً إلى ذلك، وكما تغير الاسم، هناك العديد من الاختلافات الجوهرية بين النسخة الهندية قيد الدراسة والنسختين الفارسية والعربية، لا سيما عدم الالتزام بالواقعية التي اتسمت بها نسخ الجزري المصورة، واختلاف النهج في الرسوم الهندسية للجهاز الشهير من مخطوطة الحيل، وهي ساعة الفيل. تُفند الدراسة الوصفية التحليلية كثرة المشاركين في تنفيذ الرسوم، وتُقدم أدلة من خلال الأسلوب الفني المُطبَّق.

الكلمات الدالة: التصوير الهندي الإسلامي، مخطوطات، فارسية، ساعات، عجائب

Significance

The paintings of *Aja'ib al-Sana'I* (Wonders of Craftsmanship) have motivated studying this manuscript, which was one of the earliest versions of *al-ḥiyal al-handasīyah* (Knowledge of Ingenious Mechanical Devices) translated into Persian in India. They showcase the local traits of the Indo-Islamic painting during the Khalji Dynasty in Malwa. Despite the significant similarities between the Persian and Arabic versions of *al-ḥiyal al-handasīyah*, this manuscript seems locally produced and directed toward the general public, unlike the Persian artistic painting traditions widespread across many regions in northern India during the Sultans era.

Justifications of the Study

The topic was selected to investigate the artistic styles in the paintings of devices and machines, a rare topic in local Indian paintings, especially the Islamic era of Northern India. I was particularly motivated to conduct this study because of my previous research on the Arabic versions of *al-Ḥiyal* by Ibn Musa and Al-Jazarī, entitled "The Art of Moving Statues in the Islamic Illustrated Manuscripts (Arabic Manuscripts)¹,"

¹ Some recent studies were conducted on the topic, such as :

Gupta Vivek, Routes of Translation: Connected Book Histories and al-Jazari's Robotic Wonders from the Mamluks to Mandu, South Asian Studies 2023, 39:2, 207-228; Khorasanizadeh Mohammad. A Survey of an Illustration of a Persian-Translated Manuscript of the Book "Fi Marifat al-Hiyal al-Handasiyya" (Manuscript No. 708 of the Library of Shahid Motahari University in Tehran) 2020; Ali ,Abdelaziz Ali Kayati, The relationship between Islamic applied arts and the science of Ingenious Mechanical Devices in the light of manuscripts of Musa Bin Shakir sons and Al-Jazar, Thesis, Fayoum University 2019; Ismail, Reham Said El Sayed, The Art of Moving Statues in the Islamic Illustrated Manuscripts (1) Arabic Manuscripts, The Twenty Third Conference of the Arab Archeologists, Fayoum University 2017, pp. 997-1033; Ismail, Reham Said El Sayed. Islamic Painting Schools in India before Mughol Emperors Region, PhD Dissertation, Faculty of Archeology, Cairo University 2015; Shovelton E.. Sultanate painting from the North Indian Subcontinent, PhD Dissertation, University of London 2009; De La Perrier, Eloise Brac. L'Art du Livre dans

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which clarified the similarities and differences with Arabic versions of *al-Ḥiyāl*, highlighting the importance of the advanced of Muslims in Malwa. Many new conceptualizations of mechanical devices and machines appeared in the text, and illustrations were created adjacent to the text, detailing how these machines functioned.

1-INTRODUCTION

This research handles different forms of paintings originally animated and drawn by the artist in different Persian manuscripts. The focal point is to examine the forms of paintings of human models and living beings in mechanical devices that are supposed to be moved by the forces of trickery and metaphysics in the design of the original device. The research paper aims to reveal one aspect of scientific life during the era of the Sultans in India, specifically the Malwa ², which needs further research due to the fame of India for its preoccupation with spiritual and Sufi aspects, making the study of such illustrated copies of *al-ḥiyāl* unique, including lively images that reflect a new direction in the lives of Muslims in India. It explores the artistic methods of "Indo-Islamic Painting"³ used in paintings of this version, of which only six paintings have been published⁴.

Illustrated manuscripts were so prestigious. They were exchanged as gifts with embassies between kingdoms⁵. Many versions are translated from the manuscript of *al-ḥiyāl*⁽⁶⁾ and written in Persia⁷. The first is preserved in the National Library in Paris

L'Inde des Sultanats, L'Universite Paris-Sorbonne 2008; Losty J. P.. The Art of the Book in India, the British Library 1982.

² A vast kingdom located to the north of the Deccan- region; Ālhrwy, N. A. B.. *Tbqāt Akbry*, Translated by Dr. Aḥmd 'Bd Ālqādr Ālšādly "Muslims In India From The Arab Conquest To Brotish Colonialism", Public Administration for Books, Cairo, 1995. p. 185.

³ This artistic style prevailed in several art centers of decorated manuscripts in the Sultans era in Northern India created for the public Indian Muslims, who has a unique style of paintintgs' decoration.

⁴ Losty Jeremiah P.. The Art of the Book in India, The British Library 1982; Shovelton Emily. Sultanate painting from the North Indian Subcontinent, PHD thesis, University of London 2009; Gupta Vivek. Routes of Translation: Connected Book Histories and al Jazari's Robotic Wonders from the Mamluks to Mandu, South Asian Studies 2023.

⁵ Persian spread as the language of literature, culture, and science in Central Asia and India during the Islamic rule; Al-Masry Hussein Mujib, Relations between Arabs, Persians, and Turks, Al-Dar Al-Thaqafiyah for Publishing, first edition, Cairo, 1421 AH - 2001 AD, p. 84.

⁶ In my previous study published in the 23rd Conference of Arab Archaeologists titled with: The Art of Moving Statues in the Islamic Illustrated Manuscripts (1) Arabic Manuscripts, I highlighted the importance of this topic, the literature, origin in ancient civilizations, and the transfer of industrial techniques of moving statues to the Islamic world. I also demonstrated the contributions of Muslims and the books they authored on =the topic, including methods

(no. 2477)hry t, while the second and third versions are preserved in the Šhyd M⁸ Library at the University of Tehran (707-708)⁹. There is an illustrated version called "Aja'ib Al-Sana'i" ¹⁰ (Pl.1), which is an illustrated Persian translation of the manuscript of al-Jazarī's "al-ḥiyal al-handasīyah" ¹¹ in the British Library in London¹².

2- Descriptive Study:

2-1- Description of the Manuscript

The manuscript's frontispiece indicates that the Persian manuscript was composed by "Mḥmd Bn Dāwd Šādi Abādy" rṣultan Nāin the city of Mandu of ¹³ Šāh (906-916 AH/1500-1510 AD⁽¹⁴⁾). The other two dates on the frontispiece of the

for operating devices powered by water pressure and other forces. Ismail, *The Art of Moving Statues in the Islamic Illustrated Manuscripts* (1) Arabic Manuscripts, pp. 997-1033.

⁷ Persian was the language of the court in India in the era of Afghan and Mughal, see Nehru Jawaharlal. *The Discovery Of India*, London 1951, P.126.

⁸There are about 50 Persian copies of al-Hiyal; Khorasanizadeh. *A Survey of an Illustration of a Persian-Translated Manuscript of the Book "Fi Marifat al-Hiyal al-Handasiyya"*, p.113.

⁹ Khorasanizadeh Mohammad. *Comparative Study of Illustration of three Manuscripts from the "Kitab al-Hiyal" Written by Badi'al-Zaman al-Jazari*, *Scientific Journal of Motaleate-e Tatbiqi-e Honar*, mth. 9 (17), 2019, 107-123; Khorasanizadeh. *A Survey of an Illustration of a Persian-Translated Manuscript of the Book "Fi Marifat al-Hiyal al-Handasiyya"*, (Manuscript No. 708 of the Library of Shahid Motahari University in Tehran), 2020, p.29.

¹⁰ Aja'ib (wonders) (Sing. ajab/ ajeeb) means a denial of what happens due to its rarity or the awe that overcomes a person when regarding something exceptional. In Surah Al-Kahf, Allah says "and it took its course into the sea in a strange way" (Verse 63); Ibn Manzur, *Lisan al-Arab*, Dar al-Ma'arif, Cairo, p. 2811; Al'Mu'jam al-Wasit, Arabic Language Academy, Fourth Edition, Al-Shrouk International Bookstore, Cairo, 1425 AH-2004 AD, p.584. Al-Sana'i (crafts) (Sing. Sani'ah) means everything made of goodness or charity. Its root "san'a" means "make". Allah says "And you will see the mountains and think them solid, but they shall pass away as the passing away of the clouds. The Work of Allah, Who perfected all things, verily, He is Well-Acquainted with what you do.", as evidence of craftsmanship; The Holy Quran, Surah Annaml, verse 88; Ibn Manzur, *Lisan al-Arab*, Dar al-Ma'arif, Cairo, p.2508, Al'Mu'jam al-Wasit, Arabic Language Academy, Fourth Edition, Al-Shrouk International Bookstore, Cairo2004, p.526.

¹¹ Ālǧzry, A. E. Ālǧām' Byn Āl'Lm Wāl'Ml Ālnāf' Fy šnā' Ālḥyl, edited by Ahmed Yusuf Al-Hassan, *History of Technology* (20), Higher Institute for the History of Arabic Science, Aleppo University1979.

¹² The British Library was contacted to obtain a digital copy of the manuscript "Aja'ib Al-Sana'i" for study and publication In Feb 21, 2018.

¹³ The scribe "Mḥmwd Bn Dāwd Bn Mḥmd Bn Mḥmwd Šādy Bādy" was a translator and scribe of the other three Persian copies of al-Hiyal preserved in Iran. See Khorasanizadeh. *A Survey of an Illustration of a Persian-Translated Manuscript of the Book "Fi Marifat al-Hiyal al-Handasiyya"*, p.113.

¹⁴Jeremiah P. Losty: *The Art of the Book in India*, The British Library 1982, p.68

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manuscript are found: the first is "sixteen Sha'ban" without a year, next to which is a note written by the translator that the copy is dated the fourth of Shawwal (914 AH-1509 AD). Significantly, the translation between Arabic and Persian is a strong and essential link between Arabs and Persians. Translated works must be "specific to its origin and new to the one for whom it is translated"⁽¹⁵⁾. The name of the translated manuscript refers to those wonders that the Arabs made, as highlighted by Al-Sana'i (crafts) ⁽¹⁶⁾. The translator is an inhabitant of the city of joy or "Shadiabad" or Mandu, the capital of Malwa, called "Mḥmwd Bn Dāwd Bn Mḥmd Bn Mḥmwd Šādy Bādy" ⁽¹⁷⁾ during the Khalji dynasty¹⁸. The work is preserved in the British Library, with accession number (Or. 13718). The copy written in Mandu, the capital of Malwa, is in Persian in the Naskh script, contains 191 folios, and the page size is (22x14.5 cm). The paper is light, the page has 21 lines, and the binding is modern. This copy is decorated with 159 technical diagrams of clocks, drinking vessels, liquid containers, fountains, and machines for rising water¹⁹.

2.2: The following section is a descriptive study of a collection of panels in the manuscript:

Plate No. (1): Frontispiece

Page No.: OR-13718-002

Publication: First Publishing

The frontispiece text contains a preface after which "Šādy Abādy" identifies himself as the author, and his patron is Sultan Nasir Shah Khalji in Mandu. He explains the six major types of mechanical arts. The first page is modern, illustrating that a later scribe wrote a new text for that page. After the end of the text, a table of letters describes parts of the devices in the manuscript's plates and a glossary, which is not found in al-Jazari's manuscript²⁰.

¹⁵ Almasry, Connections between the Arabs, Persians, and Turks, p.99.

¹⁶ Almasry, Connections between the Arabs, Persians, and Turks, p.22.

¹⁷ For further details on his life and works, see Turki Muhammad Ali, Iskandari Ali Asghar. Shadi Abadi and his commentary on Anuri's Divan, Letter Academy, Special Issue of the subcontinent, winter, No. 3. 2014. 115-132 (In Persian).

¹⁸ In the Khalji Dynasty, Malwa was rival to Shiraz and Samarkand, with the capital "Mandu" called "Shah Abadi", then renamed "Shadi Abad" meaning "City of Joy". Sultan Ghiyath al-Din and his son Nasir al-Din Khalji showed great interest in amusement and fun; Al-Harawi, Tabaqat Akbari, p. 234.

¹⁹ Gupta, Routes of Translation: Connected Book Histories and al-Jazari's Robotic Wonders from the Mamluks to Mandu, p. 209.

²⁰ Gupta, Routes of Translation: Connected Book Histories and al-Jazari's Robotic Wonders from the Mamluks to Mandu, 209.

The text begins with the Basmalah [In the name of Allah] in black ink, accompanied by (Rabī yassīr wtm̄ bi-al-khayr) [My Lord, make it easy and complete it with goodness] on both sides. Above the Basmalah is a rectangular area spanning the width of the page containing a Bukhariyya decoration of a lobbed oval shape in the middle with repetitions of half-Bukhariyya forms bordering the right and left sides of the rectangular area. A brown pen outlines the Bukhariyya, while the internal space is filled with red floral motifs of leafy and flowering branches on a gilded background and a blue background in the center. The rectangular Bukhariyya area is framed with a wide golden frame with short linear decorations and topped with five lobbed arches with floral motifs and an alternating blue and pink background. These decorations were added later because no page of the manuscript contains similar gold, blue, brown, and pink floral and geometric motifs.

Plate No. (2): Water Clock

Page No.: OR-13718-006

Publication: Losty, the Art of the Book in India, p.68.

Description: The representation of the clock is similar, in terms of the general external form and formations, to the model presented by Al-Jazarī, except that the artistic formations drawn are utterly different, in terms of artistic style, highlighting a style prevalent in Mandu, belonging to the Arab school in Islamic painting and strongly influenced by Jain artistic influences⁽²¹⁾.

The artist's precision in drawing the geometric components of the device can be seen, but the addition of people reflects a distinctive style for the city of Mandu, as the musical choir drawn below the arcade has colorless clothes and turbans (Fig. 3). The artist's color of the skin is light brown, a characteristic of India. The middle man sits on the surface of the Indian drum, which is distinguished by the presence of two surfaces. On his right and left are four men; one has a beard and holds a drum with two surfaces, another holds cymbals, and two blow the long trumpet. The two pots are topped by two eagles (Fig. 20). The upper composition is semicircular, divided into five parts by double lines, each of which has symbols of the zodiac signs (Fig. 1) according to the prevalent artistic style in the era of the sultans in India. Below are twelve doors expressing the number of hours determined by the device. Notably, the artist did not care to draw a man coming out of any of the doors as in the original clock. Most of his interest was in the specific external form of the water clock with its parts previously defined by al- Jazarī, but the implementation of the human drawings was inspired by Indian iconography.

Plate No. (3): Work of the moving media of the hands

²¹ Ismail. Islamic Painting Schools in India before Mughol Emperors Region, p. 620.

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Page No.: OR-13718-024

Publication: First Publishing

Description: The drawing illustrates the work of the components of the water clock and the production of sounds by drawing the weight, the broken steering column connected to the falcon's head and legs, and the weight at the bottom of the cup connected to the end of the back weight. Below the composition is a young boy drawn as if he was flying, holding a bow-shaped instrument, perhaps the stick used to beat the tambourine (Fig. 4). Attached to the right hand is a wooden joint that shows how the device is connected to the human body so that over time he can move the hand with the stick to beat the drum, announcing the passage of 12 hours. Remarkably, there was a successful attempt to draw the bird with long feathers at the end of its tail (Fig. 21), but the failure was manifested in drawing the little boy with the wide yellow clothes and brown skin like a doll.

Plate No. (4): Rotation of the images of the horoscopes in the dome above the water clock formation

Page No.: OR-13718-045

Publication: First Publishing

Description: The artist drew a huge circle with several smaller circles inside it. The space of the inner large circle is occupied by images of the twelve horoscopes without order. Some are repeated, such as Virgo, Capricorn, and Aries (Fig. 2). The artist tried to show the connection of the images to the outer and inner circles to illustrate their movement.

Plate No. (5): Dragon and Falcon Clock

Page No.: OR-13718-057

Publication: First Publishing

Description: Although the "dragon" element was one of the Chinese artistic influences and India was closer to China than Persia, the drawing of the dragon appeared different from the usual Chinese form in illustrated Persian manuscripts. Here, it has a head resembling a rooster's head, a crest with a pointed mouth resembling a beak, wide almond-shaped eyes, and a serpentine body with scales like fish scales (Fig. 26). The head is below the bird's mouth while the body is curled up inside the clock face. However, the drawing does not care about the precise implementation of the real design of the elephant clock. Additionally, there is a clear reduction in its distinctive components with the elephant and the man at the top. The artist's influence by his environment is clear in implementing the components.

Plate No. (6): Human model above the dragon clock

Page No.: OR-13718-059

Publication: First Publishing

Description: The artist drew a man above the top of a modified plant leaf crowning the top of a dome and made a rectangular frame as if it were a house (Fig.7). It is the complementary part of the clock from above to throw the weight into the mouth of the dragon in the plates of Al-Jazarī's manuscript.

Plate No. (7): Explanation of the method of moving the dragon

Page No.: OR-13718-061

Publication: First Publishing

Description: A single drawing of the dragon design separate from the device, showing how the dragon's body works by connecting the lower neck of the body to the end of the tail (Fig. 27). Another detailed drawing of a three-part moving wooden structure reveals the internal structure of the dragon's body under the scale decoration.

Plate No. (8): A device for measuring the amount of blood

Page No.: OR-13718-073

Publication: First Publishing

Description: In this plate, the artist shows a very simple design for how to make a device for measuring the amount of blood drawn from a patient. A tank with a pulley is attached to it with two measuring columns, a weight hanging on the right, and a float on the other. The pulley is attached to the bottom of a board divided into twelve units. The more blood is poured into the tank, the float rises, and the pulley moves to record the amount of blood in the tank. The man on top of the measuring board points with a pen in his hand to the number that shows the amount of blood drawn (Fig. 12).

Plate No. (9): Explanation of how the elephant clock works

Page No.: OR-13718-075

Publication: First Publishing

Description: The artist chooses an angle different from the one popular in the Arabic and Persian versions of Al-Jazarī's manuscript. Moreover, drawing the mahout riding the elephant's back in a facing position is new and different from the famous side position in the Arabic versions²². In addition to drawing the elephant owner's face not in the dark color of the skin, which is prevalent in Arabic versions, the face is of one of the famous gods in India; besides drawing the body of the elephant's legs below the elephant owner (Fig. 17), and drawing a vertical section of the way the elephant owner's body works, which makes a sound after the tank inside the elephant's body is filled with water, resulting in the tank descending to the bottom and the chains tied to the scepter. The hook held by the elephant owner's hands is tightened, which leads

²² Ismail. The Art of Moving Statues in the Islamic Illustrated Manuscripts (1) Arabic Manuscripts, p. 1010, pl. 11(أ - ب - ج - د).

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during the day to the scepter and hook colliding with the elephant's outer body, which makes a sound.

Plate (10): Human model of a prince in a roshan (window)

Page Number: OR-13718-090

Publication: Gupta, Routes of Translation, Fig. 8.

Description: The artist drew a rectangular vertical geometric composition (window or roshan) and divided it into three parts, making the middle the widest. A seat has long, thin legs on which a man sits in the Arab style (cross-legged), extending both hands. The head is covered by a turban with vertical ribs resembling lobes, and the man wears a tight robe from which the arms and chest muscles are prominent. However, the artist did not color the human model and defined the shape of the wide eyes and the direction of the pupil to the right, in addition to the plump cheeks, the large nose, and the small mouth. The man peers through an opening with a pointed arch; the artist colored the arch's spandrel in red, giving depth to the prince's seat beneath the arch. Perhaps the same style was popularly called the "Jahrokha," which would later appear frequently in Mughal paintings in India²³. Above the lobed arch, the artist drew an opening with a semicircular arch, which expressed three successive semi-circles in red lines. This gradual sequence made it a form of embodiment within the composition. The arch's spandrel was colored yellow. In the space below the window, the artist drew an onion-shaped yellow dome surrounded by a white frame and blue, red, and green.

Plate No. (11-12): Peacocks' clock

Page No.: OR-13718-100 / OR-13718-103

Publication: First Publishing

Description: It is a device used to measure time, consisting of three peacocks, each in a pointed-lobed niche, and two peacocks facing each other inside a poly-lobed arch. The artist did not show the connection between the peacocks' legs and the device's pipes at the bottom, especially in the second part of the painting, which includes the pair of peacocks. Nevertheless, the drawing of the pair of peacocks was successful, especially in drawing the legs, and he did not succeed in showing the diversity of the peacock's feathers in reality, as each peacock came in only one color. There was also a severe failure in drawing the shape of the huge peacock, so the drawing almost came close to the shape of a duck if not for the tail feathers.

²³ Fathe Saleh. The Jahrokha Drawings and Its Rituals in Light of Samples of Painting Manuscripts and Albums of the Mughal Indian School (932 1274 AH/ 1526- 1858 AD)"An Archaeological Artistic Study", BCPS, Volume 38-Issue-1, pp. 827-882.

The drawing of the peacock clock is divided into two panels: the first (Pl. 11) is located in the introductory space of the clock. We see three peacocks, each in the middle of a niche in opposite and deliberate positions, with the beak pointing downwards (Fig. 23). The panel in the lower third of the page is an integral part of the text, while the other panel, which contains the method of operating the device through its vertical section, occupies an entire page (Pl. 12 – Fig. 24).

I argue that the scribe may not be the same as the artist and that he was not familiar with the implementation of these devices and tools. His main concern was to explain the context with paintings as much as possible for the popular.

Plate No. (13): The Swordsman's Clock and the Candle

Page No.: OR-13718-2-008

Publication: First Publishing

Description: It is a device that measures time by lighting a candle. The more the candle burns and its weight decreases, the more weight increases on the other side, which leads to moving the steering column and the swordsman to remove the burned part with his sword and move the monkey (Fig. 16). The artist failed to show the correct partial connections between the parts of the machine, confirming that the artist was not well familiar with the parts of the machine and was merely translating the paintings in his artistic style prevailing in the city of Mandu, the capital of the Malwa region in India.

Plate No. (14): Ruling on drinks gatherings

Page No.: OR-13718-2-018

Publication: First Publishing

Description: It is an entertaining device used to show some humor to the attendees at drinks gatherings. In the original configuration of the device, it is placed in the middle of the hall and is operated. Thus, its upper part rotates until it stops, pointing to one of the attendees, and the musical choir begins to play, so the cupbearer rushes to him to fill his cup. Here, we see only the musical choir without the upper part (Fig. 5 - A). The artist drew the geometric shape of a tower consisting of a rectangular base with legs connected from below, topped by a square-shaped formation with a rectangular opening arched with a tri-foiled arch, the middle of which is pointed. Below sits a man wearing a tight robe and a ribbed skullcap over his head. It is connected to the lower part of the tower through a protruding part below it (Fig. 5-B), topped by a rectangular formation covered with a dome. On the left side of this tower, a vertical drawing of the musical choir appears, from bottom to top, a stringed instrument player, a flute player, a drummer, and a player holding a copper rose that may be a cymbal. Everyone sits in the Arab style (cross-legged) except for the first player, the oud player, who squats. They all look to the left. They wear white, red, and

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green cloaks, and the ribbed skullcap covers their heads. On their feet are slippers with a triangular front. As for the facial features, the skin color varies from yellow to light brown and is tinged with red. Furthermore, they have round faces, wide eyes, long prominent noses, and long thin mustaches.

Plate No. (15): Water Boat (Kushti)

Page No.: OR-13718-2-024

Publication: Shovelton, Sultanate painting, pl. 2.154.

Description: An entertaining machine whistles every half hour as water replaces air inside a lower basin. Hence, the air finds no outlet except for fine tubes connected to the bodies of a musical choir that makes a sound when air flows into it and is placed in a pool of water. The local features dominate the people, such as the skin, clothes, and musical instruments of the people of India. Additionally, the clothes and features are greatly similar between the musical choir and the prince sitting inside the pavilion on the right of the painting (Fig. 6). However, there is a clear defect in the drawing of the feet of the members of the musical choir, as they resemble claws ⁽²⁴⁾.

Plate No. (16): Drawing a human model next to a huge cup

Page No.: OR-13718-2-025

Publication: First Publishing

Description: The artist drew a huge cup-like vessel with a base, body, and top. Next to it is a drawing of a small human being whose hand extends to connect to the body of the cup. Behind the human model, a shape represents the industrial method of connecting human models to the steering column. The artist drew the human model as a plump, pretty girl (Fig.19), who wears loose clothes and a turban. She is barefoot with anklets.

Plate No. (17): A young man sitting while offering a mirror

Page No.: OR-13718-2-036

Publication: Gupta, Routes of Translation, Fig. 7.

Description: The artist drew a model of a human representing a device in the form of a young man sitting (cross-legged) holding a mirror in his left hand and a fork-like shape with three ends in his right hand, probably a comb. (Fig. 14). This painting is an attempt to implement the geometric details written to draw the model from the inside as a semi-sectionalized drawing of the figure and from the outside. However, it was unsuccessful due to the lack of connection between the drinking tank at the top of the head and the pipes and internal channels that come out of it with the moving arms to the right and left. The artist drew a young man with a foolish face with eyes opposite

²⁴ Shovelton. Sultanate painting, p.201.

the pupil towards a huge nose, heavy eyebrows, full cheeks, and a neck with thick flesh. Consequently, successive lines appear. Below them is the internal tank that stores the water inside. Above the head, we see the shape of the turban with successive folds.

Plate No. (18): A standing boy offers a drink

Page No.: OR-13718-2-038

Publication: Gupta, Routes of Translation, Fig. 6.

Description: The artist presented a model of a drink-serving device by a plump human model standing in a three-quarter position, wearing a vest and wide trousers with colored stripes of green and black. The green color was applied with a brush, while the black color was added with a pen and wrapped around the waist with a yellow shawl. The head is covered with a ribbed turban shaped like a flower's vegetal lobes, resembling a crown rather than a turban (Fig. 9). The artist described the shape of the weight placed inside the device as a triangle ending in an upper circle while the arms extended forward. In the right hand is the form of the tank and the opening with its lower rib extending to a tube. The left hand holds a red fish, and its steering column extends from the hand through the eye of the fish. Below is a similar fish, but it is uncolored and surrounded by inscriptions and symbols in Arabic. The face is not much different from the previous (plate No. 17), with a fat face, wide eyes, connected eyebrows, a large nose, and a small mouth. The original example of this model in a version of *Al hil Al handasya* was kept in Topkapi Saray copy (602AH-1206AD) Diar Bakr (plate. No. 35)²⁵.

Plate No. (19): Hand-washing device model

Page No.: OR-13718-2-039

Publication: First Publishing

Description: The artist painted a servant standing in a three-quarter position, offering a mirror with his right hand and a bottle of water with his left. In the original model, the device's owner opens the water tap to wash his hand, performs ablution with water running from a bottle or a container, and then closes it. The water pressure operates and replaces the air in the right arm, which extends after the device owner finishes ablution to present a towel or a mirror. The style of clothing seems to be representative of the servants and slaves inside the palaces in terms of the modesty of the clothes, which consist of a vest and wide trousers wrapped with a yellow shawl around the waist, with no jewelry. This is illustrated through the full plump bodies, as well as the faces with rectangular cheeks, large noses, wide eyes, and connected

²⁵ Ali, *The Relationship Between Islamic Applied Arts and The Science of Ingenious Mechanical Devices*, pl.111¹.

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eyebrows, with a mustache hanging over the small mouth. The artist presented the human model without shoes as if he was drawing a person, not a model of a device that provides a service to its owner (Fig.8).

Plate No. (20): Two waiters exchanging drinks

Page No.: OR-13718-2-040

Publication: Gupta, Routes of Translation, Fig. 2.

Description: The artist painted a rectangular geometric form based on two columns and covered it with a small dome ending at the top with a plant leaf. Below the dome is a tank with two containers from which several pipes emerge, extending to the sides to surround the setting. The connection with the human model was supposed to be drawn, but the artist did not care. Inside the seat sits a human model of two men, each holding a bottle and a glass and pouring for the other, surrounded by many geometric symbols written in Arabic, the language of science at that time²⁶. The two men appear to be sitting cross-legged in a three-quarter position, each wearing a robe with a triangular opening in front of the chest and belted around the waist; the right wears a blue robe, while the left wears a yellow one. A tight ribbed turban is above the forehead as if there is a ribbon tying it in front, while the faces are full, the eyes are directed to one another, and the beard is thin and neat around the full jaw (Fig. 15). The color scheme is limited, like the previous plates, to blue, yellow, red, and green, with a little color mixing in the dome in purple and above it the plant leaf in brown. The original example of this model in a version of *Al hil Al handasya* was kept in Topkapi Saray copy (602AH-1206AD) Diar Bakr (plate. No. 36)²⁷.

Plate No. (21): A maid holds a glass of drink

Page No.: OR-13718-2-042

Publication: First Publishing

Description: The artist painted a detailed illustration of the device on the right side of the page while the text occupied the rest of the space. The device is a domed box with legs, divided into three sections: the upper one is the tank covered by the dome, the middle is the largest and contains a girl standing on a wooden board with wheels, and the lower forms the legs of the box. In the original version, the image illustrates the text by creating clear links between the details of the device. Here, only a narrow opening for the drink to pour from the domed tank to the one above the girl's head is seen (Fig. 18), in addition to a string with an arrow at the end hanging in front of her hand. While the girl stands barefoot, wearing a green robe with her head covered in a

²⁶ Ismail. *Islamic Painting Schools in India before Mughol Emperors Region*, p 452.

²⁷ Ali, *The Relationship Between Islamic Applied Arts and The Science of Ingenious Mechanical Devices*, pl.113¹.

yellow cap with black lines and extending her left hand towards the arrow hanging with a thread, the wooden board with gliding wheels is a trapezoidal geometric shape with two red circles below it. The artist added some illustrative marks in the vaulted space at the top and next to the arrow hanging in front of the girl's left. The artist manifests his artistic abilities in illustrating the text but fails to show how the device works. The artist excels in painting the maid with the same distinctive skin color of Indian people, while adhering to avoiding nudity or embodiment.

Plate No. (22): A jug with a dove head

Page No.: OR-13718-2-045

Publication: First Publishing

Description: The artist painted a jug with a base, body, spout, handle, and faucet in the shape of a dove's head and neck. The artist outlined the body of the jug with a red or black line at the bottom without coloring. Most of the interior space was covered with inscriptions showing how the device is manufactured and operates. The faucet alone takes the shape of a dove's head and neck, colored in green, outlined with a red line, and given a beak (Fig. 22). It shows part of a device that relies on moving the jug.

Plate No. (23): Semi-sectionalized drawing of a drinking device in the form of a waiter

Page No.: OR-13718-2-052

Publication: First Publishing

Description: The artist painted a composition resembling a robot, which serves drinks to its master or owner, replacing the need for slaves or servants. In the original manuscript, the painting of this image and the arrangement of the text and image together are in the same format here, which suggests that the artist may have seen an original copy of Al-Jazarī's manuscripts that followed the same format. The artist painted a waiter standing in a three-quarter position, with a model of the waiter inside, which is a vertical sector illustrating how the device operates in terms of the tank at the top of the head and then the internal pipes that allow the drink to pour from top to bottom after the device owner opens the tap painted in front of the chest. The artist illustrated how the device operates in a style that is completely different from the other manuscript's figures, even in terms of dimensions that are relatively larger. However, only simple linear drawings of the composition are found, without any coloring, except for the tap, with some parts colored blue (Fig. 10).

Plate No. (24): A tap sided by a colorful peacock

Page No.: OR-13718-2-045

Publication: First Publishing

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Description: A single plate illustrates how the tap works and how the pipes are connected to it. The artist painted it by creating the shape of a conical glass that tapers from top to bottom while being headed with a rounded top that represents its key. Below, a narrow opening connects to the external pipes. The artist painted a huge peacock with colorful feathers, lowering its beak as if picking up something from the ground (Fig. 25). What is the connection between the tap and the peacock painting? Nothing connects them except that the tap is used to release the liquids trapped inside the tank. In the original version, peacock is pouring water from its beak. The original example of this model in a version of *Al hil Al handasya* was kept in Topkapi Saray copy (602AH-1206AD) Diar Bakr (plate. No. 37)²⁸.

Plate No. (25): A device measuring the amount of blood taken from a patient

Page No.: OR-13718-2-054

Publication: First Publishing

Description: The artist painted a form identical to the image in the original version, confirming his familiarity with the original depicted in the designs of the devices. The device operates by displacing the liquid, "the blood drawn from a patient," in a cavity of a lower tank with a float. The more the amount of liquid increases, the higher the float rises, moving a drive shaft that connects it from above to a weight hanging in the cavity of an adjacent tank. This movement between the two tanks moves the drive shaft that is connected to an upper indicator, which moves a human model carrying another indicator on the surface of the device, showing the amount of blood drawn, measured in two dirhams. Notably, the artist added many symbols next to the form to identify the dimensions of the device's components. The artist neglected to paint the tank in the middle into which the liquid directly enters, along with neglecting to depict the link holding the moving drive shaft between the two tanks and the human model above the device. Additionally, the artist overlooked other details, including painting the human model in a relatively small size and without a pointer, illustrating it with the lines that defined its size and clothes, almost without a head, as if the one who executed the geometric designs was perhaps the translator and scribe "Šādy Abādy," while the one who painted and added the human drawings and living creatures was probably another artist from India.

Plate No. (26): A vertical sector of water supply device

Page No.: OR-13718-2-059

Publication: First Publishing

²⁸ Ali, *The Relationship Between Islamic Applied Arts and The Science of Ingenious Mechanical Devices*, pl.118¹.

Description: The device consists of three layers. The first represents the top of the device, which is colored yellow with the word "gold" written in Arabic. A red ball is drawn on the external surface, from which two tubes extend inward, connected by a thick thread that passes through the second layer of the device, with four columns and a red basin. The thick thread passes through the columns to reach a red spherical weight drawn above the surface of the water that fills the lower tank, a basin through which water enters, colored in red. The lower layer is the base, consisting of a tank that contains blue liquid representing water, which was expressed by coloring its area in blue and then creating wavy black horizontal lines. A yellow tap highlights the base edge with a yellow spout and a green handle. Above the whole form, the artist added a human model sitting cross-legged, extending hands to the right and left in a facing position (Fig. 13). The human model wears a red robe with a triangular opening on the chest, closed under his left arm, and is girded with a belt at the waist. The head is covered with a turban with vertical folds defined by a thick line. The legs look like claws (water boat, plate No.15), expressing the shape of a triangular-toed slipper.

Plate No. (27): A boat-shaped alarm that beeps after a period

Page No.: OR-13718-116

Publication: First Publishing

Description: The artist painted a description of the text on the side margin of the page, using the shape of a rectangular jug containing two candles on the sides and a limited amount of water. At the top stands a human figure (a man) holding an oar in his right hand while holding and blowing a flute in his left (Fig. 11).

3-Analytical study: The images of the manuscript "Aja'ib al-Sana'i'" are characterized by the following features.

3.1- Image dimensions:

This scientific manuscript contains paintings with a highly significant illustrative function that is of no less importance than the text itself. Such paintings reflect the original initial conception of the device as perceived by its inventor and innovator, Ismail al-Jazarī. Therefore, the paintings and the figures are all cohesive with the text, as they are situated at the top of the text (plate No.13), occupy the right margin (plate No.10), or occupy the left margin (plate No. 27). The artist might prefer to add them as illustrations. They were added before the inscriptions due to the prominence of the human figures within two lines, while the words were shortened and completed in the following lines. Another format showed how the painting occupied the lower space (plates No.18-19), a space within the text (plates No.16-17), or the entire page space (plates No. 2-12-15).

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3.2- Colors: The color scheme was limited to basic bright colors, while the artist resorted frequently to the addition of color shades. The colors used were red, yellow, black, light, and dark blue. However, some paintings were not colored, as if the manuscript was not completed. For example, in (plate No. 2) of the water clock, no significant coloring of the choir's clothes was found. In (plate No.11), the peacocks were painted without coloring. The same applied to (plate No. 23), in which the coloring was limited to the tap in front of the waiter's chest.

The coloring style revealed that the artist utilized outlined contour lines of the external frame, followed by coloring. He employed a few colors, like the Indo-Islamic style, as seen in the paintings of Hamzanama and Chandayana. This artistic feature was important in the oldest schools of Indian paintings in Pala²⁹. Red and brown ink were used with touches of watercolors in green, blue, pink, brown, and red. Moreover, the frontispiece was modern³⁰.

The plates lacked a colorful background based on the original paper's color. Despite the great similarity with the Mamluk version in Deasqus(715AH-1315AD)³¹, we stress the direct copying from a Persian version copied by Šādy Bādy in Shiraz. He probably implemented the geometric schemes of devices in red or brown ink, and another Indian artist added people and animals and decorated clothes following the dominant Indo-Islamic style of non-royal workshops in Northern India.

3.3-Who was the artist?

Despite the large number of images in the manuscript, only a few contained figures of people, which helped identify the artistic style of drawing people, which became clear through two main styles. The first was rarely seen, which was the elegant drawing of the character, such as (plate No. 23, fig.10), that resembled the paintings of the "N'Mt Nāmh" manuscript³², despite its lack of skill. The second style was the most dominant in appearance and the simplest in terms of approach. The style of clothing and turbans resembled the drawing of Muslims in Jain manuscripts⁽³³⁾. Moreover, several differences appeared between the Ālšyrāzy versions and the

²⁹ Ismail. Reham Said El Sayed. Artistic archaeological study for a set of selected miniatures from the Chandayana manuscript which is preserved in the State Library in Berlin (MS. or. fol. 3014), JARCH, Volume 16-Issue 27, P. 650.

³⁰ Gupta, Routes of Translation: Connected Book Histories and al-Jazari's Robotic Wonders from the Mamluks to Mandu,p.212

³¹ Ismail. The Art of Moving Statues in the Islamic Illustrated Manuscripts (1) Arabic Manuscripts, p. 1005.

³² Ismail. Islamic Painting Schools in India before Mughol Emperors Region, p. 226.

³³ Losty. The Art of the book In India, p.68

Mandavi version, such as the scarcity of decorations in the people's clothing, in addition to the drawing of the eyes and facial features that were similar to the character of Šāhy in the genetic depiction and ḥmza Nāma, also the absence of halos⁽³⁴⁾.

To conclude, more than one person decorated the manuscript. The first was the scribe (Šādy Ābādy), who executed the geometric drawings and added some human drawings, especially (plates No. 23, Fig.10). As for the other Indian artist, he probably worked in decorating manuscripts, and Šādy Ābādy recruited him to complete the designs of the other living creatures in Indo-Islamic style, representing models of machine parts. Thus, the artistic heritage of the Sultans School was evident in the drawings, especially the human ones, where men and women were personified, and in the drawings of animals, with the integration between man and animal (plates No. 9-13, Fig. 16-17), as was customary in the Indian environment, where nature was respected and sanctified. It was also manifested in the drawing of birds, especially peacocks, with high precision in drawing all their details (plates No. 11-12-24, Fig. 23-24-25).

Although this important manuscript was copied and illustrated under the patronage of Nasser Shah Khalji in Malwa, why did this copy "*Aja'ib al-Sana'i*" not carry the same artistic features that appeared in the other royal manuscripts there, like Ni'mat Nama and Bostan Sad'I manuscripts (both were executed under patronage of Nasser Shah Khalji)³⁵? The suggested answer is: Shadi Abadi executed "*Aja'ib al-Sana'i*" under royal patronage for the Indian artisans that would be understood and apply those mechanic designs.

3.4- Drawings of geometric devices: Regarding the model of the original manuscript from which the scribe and artist copied, did it have pictures or just an explanation? If it had pictures, did it have Arab or Persian features? Did it actually come from Iran to India? Did it come directly from the Arab lands to India with scholars, writers, and calligraphers who came from various cities intending to visit the Indian courts because of their good reputation for welcoming and honoring guests, especially those who have experience in literature, art, and science?

The drawings showed the great similarity between this copy and other copies of the Arabic and Persian manuscript of *Al-ḥiyal*, in addition to the precise drawings of the semi-sectionalized devices. Therefore, the copy under study was probably copied from a Persian model that contained only geometric drawings without living creatures (plate No. 33). Additionally, the artist transferred the drawings of devices and tricks as

³⁴Shovelton. Sultanate painting from the North Indian Subcontinent, p.201.

³⁵ Ismail. Islamic Painting Schools in India before Mughol Emperors Region, p. 226- 271.

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they were and added only the shapes of living creatures to them. First, many Arabic copies of the manuscript of Al-ḥiyāl only contained geometric drawings of devices and lacked drawings of living beings. Second, the Indian character was clear in drawing living beings.

3.5- Human drawings: The artist drew figures as if they were closer to children's toys and not real human figures, as evident from the small anatomical proportions that the artist relied on in the implementation despite the spacious space for him (plate No.16, Fig.19), and the possibility of making the drawings larger in an appropriate manner in terms of anatomical proportions. However, the artist, who was not the same as the (scribe), preferred that the geometric elements of devices occupy a larger space than the arms and moving elements. In addition, the artistic style of drawing people highlighted Indian nature, with dark skin and Arab clothing, and the usual appearance of Muslims in Indian court.

The moving figures in the paintings of this edition revealed the artist's knowledge and familiarity with local Indian fashion styles and forms, which did not appear much in the Mandu manuscript illustrations. They were influenced by traditional Iranian painting, especially the style of Shiraz in the Timurid era.

The simple drawings of individuals and the whole style were clear and had a strong presence of the realism necessary in education. This same educational attitude diverted the images from what they should have and limited the view of the subject to considering it merely clear educational shapes and drawings.

The artist added a touch of sarcasm and made the human model presented for fun so that it would perform a function in terms of form, which served drink. In terms of content, it would add a spirit of sarcasm and fun to the machine's owner.

3.6. Drawings of birds and animals: The artist drew them as shown in Indian nature or as they were inherited from him from the iconographic elements and not as they appeared in the original versions of Al-Jazarī manuscript that we studied before³⁶. For instance, the eagle in the water clock (plate No.2), in a sideways position, spreads its wings, crouches on the ground, and extends its beak to the side, while it is supposed in the text that this beak extends to drop the weight into the cup drawn in front of it after a period.

3.6.1. Peacock: (Fig.14-15)

³⁶ Ismail. *The Art of Moving Statues in the Islamic Illustrated Manuscripts* (1) Arabic Manuscripts, p. 1007, Plate 7, Fig. 10.

It is the most beautiful bird; it has many colors that match each other, including green, yellow, and golden feathers. It is believed that it repels insects⁽³⁷⁾, so it is one of the birds that are widely spread in the Indian regions. It is one of the birds that the people of India love and let it go freely. The peacock is associated with the Indian gods. It is argued that it is the king of the bird kingdom, and people use its feathers to fend off and repel evil spirits⁽³⁸⁾. It appeared in (plate No.11) very similar to the drawing of the eagle in (plate No.2) in terms of the way it was drawn in a side position. It crouched on the ground while its beak extended sideways downwards. It was drawn in a relatively large size (plate No.2) next to a tab as if it was easy to use on the peacock.

3.6.2. Dragon: (Fig. 16-17)

It is a mythical creature that humans have told stories and myths about. Different descriptions and visions have been reported about it, such as considering it a terrifying sea creature or a type of huge snake ⁽³⁹⁾. This is the typical form of the dragon drawn in a painting from Hamza Nāmh⁴⁰. Perhaps the artist Hamza Nāmh did not see the model of the dragon drawing in Persian manuscripts, so the drawing of the dragon as it is in the popular heritage resembles the water snake (Plates No. 4-5), where the long black body is twisted around a round head. The drawing of the dragon in the form of a snake was a symbol of evil in ancient beliefs and myths⁽⁴¹⁾. The artist drew the dragon with a serpentine body like a snake, with scales resembling a crocodile's body, and a head resembling a rooster's head, which was different from the same approach in the Arab and Persian painting to the dragon's form.

3.6.3. Drawing the elephant and the mahout

In one unit in a facing position with great mastery of the rules of perspective (plate No. 6) is a new approach that no one has ever done before. Only the two front feet of the elephant and its huge white hooves appear, and the rest of the body disappears completely behind the lioness with its oval face and long nose that resembles an elephant's trunk and drooping ears (Fig. 4). The artist drew below a semi-sectionalized drawing of the way water descends to the lower tank in the original

³⁷ Al-Qazwini Zakariyya. *Ajaeb Al-Makhloqat*, Al-Aalami Publications Foundation, Beirut, First Edition, 1421 AH- 2000 AD. p. 349.

³⁸ Thaqafa Alhend, *Superstitious Beliefs in Indian Savage Tribes*, Volume III, Issue Three, June 1953. p. 22-26.

³⁹ Ramadan, Hussein Mostafa Hussein. *Jozahr in Islamic Art*, Journal of Faculty of Archaeology, Volum.14, Cairo 2009, p.126.

⁴⁰ Ismail. *Islamic Painting Schools in India before Mughol Emperors Region*, p.47.

⁴¹ Khatari Earabi. *The Mythological Construction Of The Popular Biography*, Ain for Human and Social Studies and research, First Edition , Giza, 1430AH-2009AD,P. 81.

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device from top to bottom, in which the elephant's huge body worked as a lower tank for water that descended into it after a specific period, which caused the float inside it to rise and the elephant to descend with its hammer over the elephant's head⁴². The artist reduced all of that process to the appearance of the mahout holding the hammer and fork, with its legs open in the shape of an arch, indicating that it is straddling the elephant, with the tank and the elephant's legs below it.

The elephant and the mahout plate (plate No. 6) and the dragon and falcon clock plate (plate No. 4) are considered the most distinctive features of this manuscript. They are a new approach, neither in Persian nor Arabic versions⁴³. They are separate parts of the same device, which is the elephant clock.

3.6.4 As for the drawing of the monkey, it was naked with thick hair but with the head of an animal, similar to the drawing of demons in Islamic paintings in Central Asia (plate No. 12 – Fig. 9). It was similar to the example in Topkapi Saray copy of *Al hil Al handasya* (602AH-1206AD) Diar Bakr (plate. No. 34)⁴⁴.

3.7. Artistic heritage: India has been famous throughout the ages for its involvement in spiritual and mystical aspects and contemplation of the forces of nature. All religions and sects that believe in the transmigration of souls have been involved. They have achieved extremism and avoided killing all living beings. On the other hand, there was a great desire for creativity in geometric and mechanical aspects as India became one of the first countries exporting technology.

Studying the illustrations of “Ajaib Al Sanaii” manuscript revealed that it was a rare copy by a scribe who was not the only artist who executed all the shapes and illustrations provided with living creatures within the text. The illustrations were distinguished by extreme modesty in colors and drawing. However, they reflected an aspect of the artistic trends of painting in Mandu, the capital of the Malwa region, which received strong cultural currents from Iran, as manifested in many manuscripts decorated with paintings. This scientific manuscript continues to be an important aspect of cultural life in Mandu.

⁴² Ismail. *The Art of Moving Statues in the Islamic Illustrated Manuscripts* (1) Arabic Manuscripts, p. 1010, Plate 11, Fig. 13.

⁴³For more about The Elephant clock in Persian Versions Of “Automata’ Al Jazri” manuscript see;

Khorasanizadeh Mohammad, Ghazizadeh Khashayar ,Morteza Afshari , Abedini Ahmad. "Elephant Water Clock" Representation in the Manuscripts of the Books of Badi' al-Zaman Jazari, *Theoretical Foundations of Visual Arts* No. 13, Spring-Summer 2022.

⁴⁴ Ali, *The Relationship Between Islamic Applied Arts and The Science of Ingenious Mechanical Devices*, pl.103.

The manuscript was copied and decorated in non-royal painting workshops in Mandu, as highlighted by the artistic style evident in the depiction of figures with Indian features and the modeling of body muscles, especially in plate number 10. It highlighted the painting style in one of the oldest translated Sufi manuscripts in northern India, 'Chandayana,' which was copied in the middle of the 9th H/ 15th AD century during the Lodi Dynasty in the reign of Sultan Iskandar Lodi in Delhi. It is preserved in the State Library in Berlin (MS. or. fol. 3014). It is also the same ⁴⁵ artistic style in the paintings of the Hamzanama manuscript, which features the style of Delhi and Agra and was copied around (9AH-1450 AD). It is also preserved in the State Library in Berlin (Or. Fol. 4181). In addition, the color scheme for all the ⁴⁶ paintings was extremely limited, which was characteristic of the paintings in the Hamzanama manuscript (Pl.28)⁴⁷ and the Chandayana manuscript (Pl.29) ⁴⁸.

Jain manuscripts in western India lacked strong Indian influences⁴⁹. For instance, the protruding eyes did not appear outside the face, but there was an intellectual unity between the types of Indian arts⁵⁰. Therefore, the touch of Indian art appeared clearly in the coloring of people and drawing of the colored peacock, especially the elephant and the mahout (plate No.9). The plate had another artistic feature, namely the similarity with an Indian deity, which was perfectly transferred to the Indo-Islamic painting in the Sultans era⁵¹.

4-Conclusion and results:

First, the study presented Aj'aib Al-Sana'i's manuscript translated into Persian in Mandu, the capital of the Malwa region, during the Khalji Dynasty. Twenty-seven paintings from the manuscript "Aj'aib Al-Sana'i" were examined, 21 paintings of which were published for the first time, in addition to 27 illustrative figures.

Second, the Arabic geometric drawings were transferred to the Persian version in Shiraz and then to Aj'aib Al-Sana'i in Mandu, indicating no major innovations, which

⁴⁵ Ismail. Artistic archaeological study for a set of selected miniatures from the Chandayana manuscript which is preserved in the State Library in Berlin, P.647.

⁴⁶ Ismail. Islamic Painting Schools in India before Mughol Emperors Region, p.47.

⁴⁷ Ismail. Islamic Painting Schools in India before Mughol Emperors Region, Pl.(1.157)

⁴⁸ Ismail. Artistic archaeological study for a set of selected miniatures from the Chandayana manuscript which is preserved in the State Library in Berlin, P. 650, Pl. 7.

⁴⁹ Gray Basil. Western Indian Painting in Sixteenth Century: The Origins of the Rajput School, The Burlington Magazine, Vol. 90 No.539 (Feb.,1948), pp. 40-43.

⁵⁰ Zhdy B. Glimpses from indian Art, Journal" Sawt El Sharq", 388 Edition, Cairo 1996, p. 12.

⁵¹ Ismail. Artistic archaeological study for a set of selected miniatures from the Chandayana manuscript, P. 650.

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asserts the superiority of Arab scientists in mechanical engineering and water metaphysics in the Middle Ages.

Third, the descriptive study proved that the translator "Mḥmwd Šādy Abādy," the scribe of the manuscript "Aj'aib Al-Sana'i," was not the only artist who painted all the forms of living creatures within the paintings of devices. Instead, another Indian artist participated in implementing the drawings of living creatures. This explains the lack of illustrating a logical connection between the painting of the living model and the drawing of the machines. However, those drawings added artistic value to the figures of the manuscript.

Fourth, the study showed the artist's ability to express the artistic features of the Indian painting school in the era of the Sultans in Mandu through the limited use of colors, tight clothes, skin tone, animal shapes, and bird drawings. He paid special attention to the prevalence of embodiment while adding some sarcastic features to the faces of his people, reflecting the spirit of his cheerful personality.

The artistic style of Aj'aib al-Sanai' miniatures were an artistic and decorative unit with decorative plates of contemporary Persian manuscripts in Delhi, e.g., Hamzanama and Chandayana in the State Library in Berlin, which were implemented in non-royal workshops. They reflected the Indo-Islamic painting in the Sultans era in Northern India.

Fifth, the original Arabic version probably did not contain drawings of living creatures but was only one of the copies of the al-al-ḥiyal manuscript that contained only geometric drawings. Therefore, there were many differences between the paintings of the manuscript of Aj'aib al-Sanai' and the paintings of the manuscript of al-ḥiyal, especially in the paintings of the elephant and its rider in the elephant clock, drinks gatherings, and hand-washing device. Differences were also traced in not adhering to the drawing and painting as in al-Jazarī's copies, which were characterized by realism, and not following the rules of accurate industrial painting to explain how the device operates, in addition to the fundamental differences in the paintings of the characters between Damascus, Shiraz, and Mandu.

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Illustrations



Figure (1) Celestial signs above the water clock



Figure (2) Twelve celestial signs



Figure (3) Musical choir



(4) Human model of the musician



(A)

Figure (5) A - Musical choir



(B)

(5) B-How to move a human maquette

(Reasercher's Work)

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Figure (6) Musical choir



Figure (7) A man upper



Figure (8) Servant



Figure (9) Servant Pavilion.



Figure (10) Servant



**Figure (11) Human being
maquette is setting
(Reasercher's Work)**



Figure (12) Flute player



Figure (13) Human being



Figure (14) Servant



**Figure (15) Two servants
maquette is setting**



Figure(16) The monkey and the sward man



**Figure(17)The mahout and
the elephant together**



Figure (18) A beautiful maid



**Figure (19)A girl stands beside
the huge cup**

(Reasercher's Work)

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Figure (20) Falcon



Figure (21) Falcon



Figure (22) Dove



Figure (23) Peacocks



Figure (24) Peacocks



Figure (25) Peacock

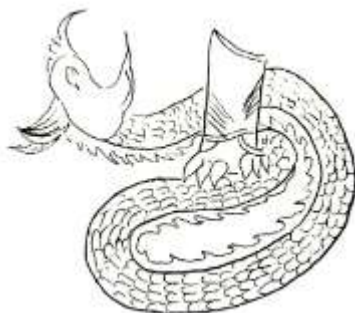


Figure (26) Dragon

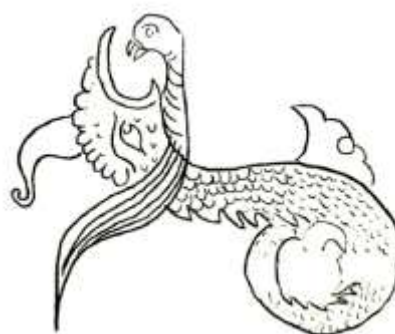


Figure (27) Dragon

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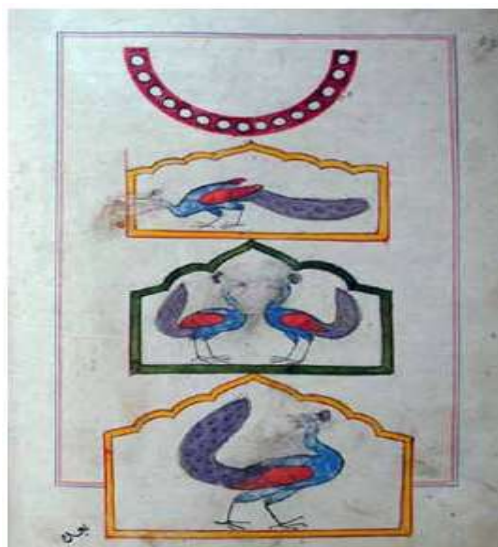
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Pl. 31. Peacock Basin, “*Fi Marifat al-Hiyal al-Handasiyya*” manuscript (No. 708), the Library of Shahid Motahari University, Shiraz (10AH-16 AD).

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**Pl. 32. Peacock Watch, “*Fi Marifat al-Hiyal al-Handasiyya*” manuscript (No. 708), the Library of Shahid Motahari University, Shiraz (10AH-16 AD).
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Pl. 35. A Servant carries A fish and cup, “*Al Hiyal al-Handasiyya*” Manuscript (No. Ahmed III, 3472), Topkapi Museum, Diar Bakr (602AH- 1206AD).

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